QP CODE: 25901911

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU- UGP (Honours)/MGU- BBA (Honours)/MGU- BCA (Honours)

Special Examinations, July 2026

(for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC)

MG3DSCMAT200–PERSPECTIVES OF MATHEMATICS

(2024 ADMISSION ONWARDS)

Scheme For Valuation

Duration: 2 Hours Maximum Marks: 70

Part A {Short Answer Questions}

Answer any 5 questions. Each question carries 2 marks

1. Find the Cartesian coordinates of the point whose polar coordinates is $(-5, \pi)$. ... **[A]/[CO1]**

   - Using $x = r\cos\theta, \quad y = r\sin\theta,$ ... 1 Mark
   - $x = (-5)\cos\pi = (-5)(-1) = 5,\ y = (-5)\sin\pi = (-5)(0) = 0.$ ... 2 Marks

   (Maximum Marks =2)

2. Find the parametric equation and parametric interval for the motion of a particle that starts at $(a, 0)$ and traces the circle $x^2 + y^2 = a^2$ once counterclockwise. ... **[U]/[CO1]**

   - Parametrization is $x = a\cos t, \quad y = a\sin t,$ where $0 \le t \le 2\pi.$ ... 2 Marks

3. If α, β, γ are the roots of the equation $8x^3 - 4x^2 + 6x - 1 = 0$, obtain the equation whose roots are $2\alpha + 1,\ 2\beta + 1,\ 2\gamma + 1.$ ... **[[A]/[CO4]**

   - Let $y = 2x + 1$. Then $x = \dfrac{y-1}{2}$.

     Substitute into the given equation: $8\left(\dfrac{y-1}{2}\right)^3 - 4\left(\dfrac{y-1}{2}\right)^2 + 6\left(\dfrac{y-1}{2}\right) - 1 = 0.$ ... 2 Marks
   - Hence the required equation is $y^3 - 4y^2 + 8y - 6 = 0.$ ... 1 Mark

   (Maximum Marks= 2)

4. If $q, r, s > 0$, show that $f(x) = x^4 + qx^2 + rx - s = 0$ has one positive, one negative and two imaginary roots. ... **[A] / [CO4]**

   - Correct proof with supporting results ... 2 Marks

5. Express the double integral $\displaystyle\int_0^2 \int_0^{\sqrt{4-y^2}} (x^2 + y^2)\, dx\, dy$ in polar coordinates. ... **[K] / [CO5]**

- To take $0 \leq r \leq 2, \qquad 0 \leq \theta \leq \frac{\pi}{2}$. ... 1 Mark
- For getting $\int_0^{\pi/2} \int_0^2 r^3 \, dr \, d\theta$. ... 2 Marks

(Maximum Marks= 2)

6. Write an equivalent integral of $\int_0^3 \int_0^{2x} dy \, dx$ with the order of integration reversed.

... **[A] / [CO5]**

- For getting $\int_0^6 \int_{y/2}^3 dx \, dy$ ... 2 Marks

7. Define spherical coordinates of a point in space.

... **[K] / [CO6]**

- The spherical coordinates of a point are denoted by (r, θ, ϕ), where
  - r is the distance from the origin,
  - θ is the angle made by the projection on the xy-plane with the positive x-axis (azimuthal angle),
  - ϕ is the angle made with the positive z-axis (polar angle).

... 2 Marks

8. Evaluate $\int_{-1}^1 \int_0^1 \int_0^2 (x + y + z) \, dx \, dy \, dz$.

... **[A] / [CO 6]**

- For getting $\int_{-1}^1 \int_0^1 \int_0^2 (x + y + z) \, dx \, dy \, dz = 6$ ... 2 Marks

[2 × 5 = 10]

Part B - Short Essay Questions
Answer any 5 questions. Each question carries 6 marks

9. Graph the set of points whose polar coordinates satisfy $-3 \leq r \leq 2, \qquad \theta = \frac{\pi}{4}$. ... **[A]/[CO1]**

- Correct graph ... 6 Marks

10. Identify the conic $x^2 - y^2 + 4x - 6y = 6$. Also find its centre, vertices and foci. ... **[A]/[CO2]**

- For getting $\frac{(x+2)^2}{1} - \frac{(y+3)^2}{1} = 1$, a hyperbola ... 3 Marks
- (a) Centre =(-2,-3)
  (b) Vertices:$(-3, -3), \ (-1, -3)$.
  (c) Foci: $(-2 - \sqrt{2}, -3), \ (-2 + \sqrt{2}, -3)$. ... 3 Marks

11. Find the equation whose roots are those of $x^4 - 2x^3 + 3x - 5 = 0$ each diminished by 2.... **[A] / [CO 4]**

- Let the new variable be $y = x - 2$. Then $x = y + 2$. ... 3 Marks
- For getting the equations as $x^4 + 6x^3 + 12x^2 + 11x + 9 = 0$ ... 3 Marks

12. How many real roots are there for $x^4 - 1 = 0$? Find all the roots. ... **[A] / [CO 4]**

- Factorize: $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x - i)(x + i)$. ... 3 Marks
- The four roots are
  $1, -1, i, -i$. ... 3 Marks

13. Find the area enclosed by the lemniscate
  $r^2 = 4\cos 2\theta$. ... **[A] / [CO 4]**

- Area of one loop is
  $A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos 2\theta \, d\theta$. ... 3 Marks
- $A = 2\int_{-\pi/4}^{\pi/4} \cos 2\theta \, d\theta = [\sin 2\theta]_{-\pi/4}^{\pi/4} = 2$. ... 3 Marks
  Hence the total area enclosed by the two loops is 4. ... 1 Mark

(Maximum Marks= 6)

14. Sketch the region bounded by the parabolas $x = y^2 - 1$, $x = 2y^2 - 2$. Express the area as an iterated integral and evaluate it. ... **[A] / [CO 5]**

- Correct graph ... 3 Marks
- For getting $A = \int_{-1}^{1} (1 - y^2)\, dy = \left[y - \frac{y^3}{3}\right]_{-1}^{1} = \frac{4}{3}$. ... 3 Marks

15. Evaluate the limits of integration r, θ and z for the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by $z = x^2 + y^2$ and below by the xy-plane. ... **[A] / [CO 6]**

- Using cylindrical coordinates, $x = r\cos\theta$, $y = r\sin\theta$.
  The cylinder gives $0 \le r \le 2$. ... 3 Marks
- The complete revolution gives $0 \le \theta \le 2\pi$. Since $z = 0$ below and $z = r^2$ above,
  $0 \le r \le 2$, $0 \le \theta \le 2\pi$, $0 \le z \le r^2$. ... 3 Marks

16. Sketch the region of integration and evaluate $\int_0^1 \int_y^1 x^2 e^{xy}\, dx\, dy$. ... **[A] / [CO 7]**

- Correct graph ... 3 Marks
- The region is $0 \le y \le 1$, $y \le x \le 1$.
  Reversing the order, $0 \le x \le 1$, $0 \le y \le x$. ... 2 Marks
- $\int_0^1 \int_0^x x^2 e^{xy}\, dy\, dx = \frac{1}{2}(e - 1) - \frac{1}{2} = \frac{e - 2}{2}$. ... 2 Marks

(Maximum Marks= 6)

[6 × 5 = 30]

Part C - Essay Questions
Answer any 3 questions Each question carries 10 marks

17. (a) Given vertices $(\pm 10, 0)$ and eccentricity $e = 0.24$ of an ellipse centered at the origin, find its standard equation. ... **[An]/[CO2]**

- For getting $b^2 = 100 - 5.76 = 94.24$. ... 3 Marks
- For finding $\dfrac{x^2}{100} + \dfrac{y^2}{94.24} = 1$. ... 2 Marks

(b) Find the eccentricity, foci and directrices of the hyperbola $9x^2 - 16y^2 = 144$.. Also graph the hyperbola.

- For getting Eccentricity $e = \dfrac{c}{a} = \dfrac{5}{4}$. ... 3 Marks
- i. The foci are $(\pm 5, 0)$.
  ii. The directrices are $x = \pm\dfrac{16}{5}$.
  iii. Correct Sketch ... 5 Marks

(Maximum Marks= 10)

18. Let the roots of $x^3 - 4x^2 + 5x - 2 = 0$ be α, β, γ. ... **[A] / [CO3]**

(a) Find $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$, $\alpha\beta\gamma$.

(b) Find $\alpha^2 + \beta^2 + \gamma^2$.

(c) Find $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma}$.

- Getting $\alpha + \beta + \gamma = 4$, $,\alpha\beta + \beta\gamma + \gamma\alpha = 5$, $,\alpha\beta\gamma = 2$. ... 4 Marks
- $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = 6$ ... 3 Marks
- $\dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma} = \dfrac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \dfrac{5}{2}$. ... 3 Marks

19. (a) Form a quartic equation having real coefficients with $1 + i$ and $2 - \sqrt{3}i$ as roots. ... **[U] / [CO3]**

- The four roots are
$$1 + i, \quad 1 - i, \quad 2 + \sqrt{3}i, \quad 2 - \sqrt{3}i.$$
... 3 Marks
- For getting the equation as $x^4 - 6x^3 + 17x^2 - 22x + 14 = 0$ ... 3 Marks

(b) Solve $x^4 - 7x^3 + 18x^2 - 22x + 12 = 0$ given that $1 + i$ is a root.

- Calculations to get roots as $x = 1 + i,\ 1 - i,\ 2,\ 3$. ... 4 Marks

20. Find $\displaystyle\iint_R (x + y)\, dx\, dy$ where R is ... **[A] / [CO5]**

(a) the region bounded by $x = 0,\ y = 0,\ x + y = 1$

- To get The limits of integration are
$$0 \le x \le 1, \qquad 0 \le y \le 1 - x.$$
... 2 Marks

- Calculations to get $\int_0^1 \int_0^{1-x} (x+y)\,dy\,dx = \frac{1}{3}$ ... 3 Marks

(b) the region bounded by the first quadrant of the circle $x^2 + y^2 = 9$

- To get the limits of integration are

$$0 \le x \le 3, \qquad 0 \le y \le \sqrt{9 - x^2}.$$

... 2 Marks

- Calculations to get

$$\iint_R (x+y)\,dx\,dy = 18.$$

... 3 Marks

21. Let D be a cube in the first octant bounded by the coordinate planes and $x = 2,\ y = 2$ and $z = 2$ and Let $f(x, y, z) = x^2 + 9$ is a function over D ... **[A] / [CO6]**

(a) Find the volume of D

- Calculations to find volume as 8 ... 5 Marks

(b) Find the average value of f over D

- Calculations to find average value of f over D is $\frac{31}{3}$ ... 5 Marks

22. Find $\iiint |xyz|\,dx\,dy\,dz$ over the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1$ using the substitution $x = au,\ y = bv,\ z = cw$.

- Calculations to get

$$\iiint_D |xyz|\,dx\,dy\,dz = \frac{a^2b^2c^2}{6}.$$

... 10 Marks

[10 × 3 = 30]